Web Programming

Semester Project Specification (Individual)

1. Project Overview

In this semester project, each student is required to **design and develop a complete dynamic website** as part of the Web Programming course.

The project is designed to assess the student's ability to:

- Apply core web technologies in a real-world context
- Build a structured, responsive, and database-driven website
- Implement server-side logic using PHP and MySQL
- Use AJAX for dynamic interactions
- Apply basic security and SEO practices
- Clearly explain design and implementation decisions

This is an **individual project** and must be developed progressively throughout the semester.

2. Project Domains (Mandatory)

Each student must choose **ONE (1)** project domain from the list below.

Domain ID	Project Domain
D1	Online Book Store
D2	Fashion / Shoe Store
D3	Restaurant / Food Ordering Website
D4	Electronics Store
D5	Online Course Platform

⚠️ Once a domain is selected and approved, it **cannot be changed**.

Domain Assignment Rule (Based on Student ID)

To reduce duplication and ensure fair distribution across project domains, **each student's project domain is assigned based on their Student ID**.

Domain Assignment Formula

Domain_ID = StudentID mod 5

Domain Mapping Table

Result (StudentID mod 5)	Assigned Domain
0	D1 – Online Book Store
1	D2 – Fashion / Shoe Store
2	D3 – Restaurant / Food Ordering Website
3	D4 – Electronics Store
4	D5 – Online Course Platform

Examples

- Student ID **20210005**
  → 20210005 mod 5 = **0**
  → **Online Book Store**
- Student ID **20210018**
  → 20210018 mod 5 = **3**
  → **Electronics Store**

Important Rules

- Students **must follow the assigned domain** based on the formula above.
- Domain changes are **not allowed**, except with explicit instructor approval.
- Submitting a project under an incorrect domain will result in:
  - **Major point deduction**, or
  - **Project rejection**, depending on severity.

3. Technical Requirements

A. Layout & Structure

The website must include the following layout components:

- **Header**
  - Website logo aligned to the left
- **Navigation Bar**
  - At minimum: Home, Products (or Items), Search, Contact
- **Main Content (Body)**
  - Layout width is flexible
  - The chosen width and layout style must be explained in the project report
- **Footer**
  - Short website information (copyright, contact details, etc.)

B. Responsive Design

- The website must be responsive and usable on:
  - Desktop
  - Tablet
  - Mobile devices
- You may use:
  - Bootstrap
  - OR custom CSS with media queries

C. Products / Items Page

Clicking **Products** (or equivalent) must display a catalog loaded from the database.

Minimum requirements:

- Products/items must be stored in **MySQL**
- Sorting must be supported (e.g., by price, name, rating)
- Implement **ONE (1)** of the following:
  - Pagination
  - Lazy loading

Students must explain **why** they chose this approach in the project report.

D. Search Function (AJAX Required)

- Implement a product/item search feature using **AJAX**
- Requirements:
  - Search results must update dynamically while typing or submitting
  - No full page reload during search
  - Users can click a result to view the item details page

E. Categories & Navigation Path (Breadcrumbs)

- Products/items must be organized into **categories**
- Breadcrumb navigation must be displayed on relevant pages
  Example:
  `Home > Products > Category > Item`

F. Store Locations & Maps Integration

- A seller or service provider may have **multiple physical locations**
- On each product/item page:
  - Display which locations have the item available (or serve it)
  - Provide a **Google Maps link** for each location

G. User Authentication System

The website must implement a basic authentication system with the following features:

- Register
- Login
- Logout
- Forgot Password

Requirements:

- Authentication using **email and password**
- Input validation (email format, password length, etc.)
- Passwords must **not** be stored in plain text
- Social login (Google / Facebook) is allowed but optional

H. Search Engine Optimization (SEO)

Apply **at least three (3)** SEO practices, such as:

- Meaningful page titles and meta descriptions
- Semantic HTML elements
- Friendly URLs (e.g., `/products/shoes`)
- Sitemap

4. Database Requirements

- The project must use **MySQL**
- At minimum, the database must include:
  - Users
  - Products / Items
  - Categories
- Database design must:
  - Use appropriate primary keys
  - Include foreign keys where relevant
  - Be clearly explained in the project report

5. Deliverables

Each student must submit the following:

1. **Source Code**
   - Organized project folder
   - Clear file and folder structure

2. **Database Files**
    - SQL file for table creation and sample data
    - ER diagram (image or PDF)
3. **Project Report (PDF, 5–10 pages)**
   The report must include:
    - Introduction and purpose of the website
    - Explanation of domain-specific design choices
    - Layout and responsive design decisions
    - Database schema explanation
    - Implementation details (AJAX, authentication, SEO)
    - Challenges faced and lessons learned
4. **Final Demonstration**
    - Live demonstration during the final lab session
    - Website must be functional

6. Evaluation & Grading (100 Points)

Component	Description	Points
Layout & Navigation	Header, navbar, body, footer	10
Responsive Design	Works on desktop, tablet, mobile	10
Products / Items Page	Sorting + pagination or lazy loading	15
Search with AJAX	Dynamic search without page reload	10
Categories & Breadcrumbs	Correct structure and navigation	10
Store Locations & Maps	Availability + Google Maps links	10
User Authentication	Register, login, validation, forgot password	15
SEO Implementation	At least 3 techniques applied	10
Project Report	Clarity and explanation of decisions	5
Final Demonstration	Working system	5
TOTAL		**100**

7. Academic Integrity Policy

This is an **individual project**.

- Students may use tutorials, documentation, and online references
- Copying another student's project (fully or partially) is prohibited
- Projects that are substantially similar may receive a zero score

During the final demonstration or review, students may be asked to:

- Explain parts of their code

- Describe design decisions
- Make small modifications

Submitting a working website does not guarantee a passing grade if the student cannot demonstrate understanding of their own work.

8. Important Notes

- Start early and work incrementally
- Use version control (GitHub / GitLab) if possible
- Focus on **correctness, clarity, and understanding**, not complexity
- This project reflects real-world web development practices

Final Remark for Students

The goal of this project is not to build a perfect or complex product, but to demonstrate that you understand how a modern web application works from front-end to back-end.